Data Analytics

Topic: **Visualization Tool for Electric Vehicle Charge and Range Analysis**

Team Members:

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VISUALIZATION TOOL FOR ELECTRIC VEHICLE CHARGE AND RANGE ANALYSIS

Data Visualization Project Using Tableau

1. PROJECT TITLE

Electric Vehicle Charge and Range Analysis – A Data Visualization Study of EV Infrastructure, Performance and Market Distribution in India

2. PROJECT DESCRIPTION

● System Overview

This project is a data visualization system designed to analyze Electric Vehicle (EV) charging infrastructure, vehicle performance metrics, efficiency levels, pricing segmentation, and powertrain distribution in India. The system integrates cleaned datasets into an interactive Tableau dashboard for analytical insights.

● Problem Domain

The EV industry is growing rapidly; however, structured analytical representation of EV data such as charging availability, brand comparison, efficiency, and pricing distribution is limited. This project addresses the gap by consolidating scattered EV data into one analytical dashboard.

● Core Technology Used

- Tableau (Dashboard Development)
- Microsoft Excel (Data Cleaning & Transformation)
- OpenStreetMap (Geographic Mapping)

● Key Capabilities

- Charging station regional analysis
- Brand & model comparison
- Performance & efficiency comparison
- Price segmentation analysis

- Powertrain distribution study

● **Target Users**

- Researchers
- EV Market Analysts
- Policy Makers
- Students
- Automobile Industry Professionals

● **Real-world Relevance**

The system supports understanding of EV infrastructure readiness, brand competitiveness, and affordability segments within India.

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3. APPLICATION SCENARIOS / USE CASES

● **Enterprise Usage**

Automobile companies can analyze competitor performance and pricing strategies.

● **End-user Usage**

Customers can understand EV pricing, efficiency, and brand comparison.

● **Educational Usage**

Students can study real-time EV infrastructure and performance data through visualization.

● **Government / Policy Usage**

Policy makers can assess charging infrastructure concentration and expansion needs.

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4. TECHNICAL ARCHITECTURE OVERVIEW

- **High-Level Architecture**

Dataset → Excel Cleaning → Tableau Dashboard → User Interaction

- **Component Interaction**

User interacts with Tableau dashboard → Filters applied → Data processed → Visual insights generated

- **Frontend Layer**

Tableau Dashboard Interface

- **Backend Layer**

Data processing through Excel & Tableau Engine

- **Database Layer**

Structured dataset stored in Excel sheets

- **External Services**

OpenStreetMap integration for geographic mapping

- **Deployment Environment**

Local system deployment via Tableau Desktop

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5. PREREQUISITES

5.1 Software Requirements

- Microsoft Excel
- Tableau Desktop
- Windows OS

5.2 Libraries / Dependencies

- Tableau Visualization Engine
- OpenStreetMap API

5.3 Hardware Requirements

- Minimum 4GB RAM
- Standard laptop/desktop

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6. PRIOR KNOWLEDGE REQUIRED

- Basic Data Analytics
- Tableau Fundamentals
- Excel Data Handling
- Data Visualization Concepts
- EV Industry Basics

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7. PROJECT OBJECTIVES

Technical Objectives

- Develop interactive dashboards
- Analyze EV charging infrastructure

Performance Objectives

- Compare efficiency and top speed
- Evaluate brand competitiveness

Deployment Objectives

- Deliver structured visualization system

Learning Outcomes

- Data preprocessing
- Dashboard design
- Insight extraction

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8. SYSTEM WORKFLOW

1. Data collection
2. Data cleaning in Excel
3. Data transformation
4. Import dataset into Tableau
5. Create visualizations
6. Build interactive dashboard
7. Generate insights

MILESTONE 1: REQUIREMENT ANALYSIS & SYSTEM DESIGN

9.1 Problem Definition

Lack of centralized EV data visualization platform.

9.2 Functional Requirements

- Display charging station distribution
- Show brand & model comparison
- Present performance metrics

9.3 Non-Functional Requirements

- Interactive
- User-friendly
- Scalable

9.4 System Design Decisions

Tableau selected for strong visualization capabilities.

9.5 Technology Stack Justification

Tableau provides advanced interactive charts and mapping integration.

MILESTONE 2: ENVIRONMENT SETUP

- Installed Tableau
- Prepared dataset in Excel
- Structured data fields

MILESTONE 3: CORE SYSTEM DEVELOPMENT

Feature 1 – Charging Infrastructure Analysis

Mapped charging stations by region and type (AC/DC).

Feature 2 – Brand & Model Distribution

Compared number of models per brand.

Feature 3 – Performance & Efficiency Analysis

Calculated average top speed and Wh/Km efficiency.

MILESTONE 4: INTEGRATION & OPTIMIZATION

- Integrated all sheets into single dashboard
- Applied filters for dynamic analysis
- Improved layout responsiveness

MILESTONE 5: TESTING & VALIDATION

Sample Test Case

Test Case ID	Input	Expected Output	Status
TC01	Select Region	Filtered charging data	Passed
TC02	Select Brand	Display brand performance	Passed

Performance Metric: Smooth filter response within seconds.

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14. DEPLOYMENT

- Tableau Desktop deployment
- Packaged dashboard file
- Shared for demonstration

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15. PROJECT STRUCTURE

```
project_root/
|
 ├── dataset/
 ├── excel_files/
 ├── tableau_dashboard/
 ├── screenshots/
└── documentation/
```

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16. RESULTS

- Identified NDMC as highest charging concentration region
- Luxury brands dominate performance
- Tata & MG offer affordable EVs
- Clear segmentation between premium & budget EVs

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17. ADVANTAGES & LIMITATIONS

Advantages

- Interactive visualization

- Clear infrastructure comparison
- Market segmentation insight

Limitations

- No real-time data
- Limited dataset scope

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18. FUTURE ENHANCEMENTS

- Real-time charging API integration
- Predictive analytics
- State-wise adoption trend analysis

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19. CONCLUSION

This project successfully demonstrates how structured data visualization can provide meaningful insights into EV infrastructure readiness, performance distribution, pricing segmentation, and market competitiveness in India.